

Machine Learning

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Lecture 17: Bayes Theorem

Important Note about sources

These slides are for my Machine Learning, Course

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References

Most of this material is from:

- *Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition*, by Daniel Jurafsky, James H. Martin; Published by Pearson Prentice Hall, 2008, ISBN 0131873210, 9780131873216
- Computational Linguistics, Slides, *Dr. Sarmad Hussain*, CLE
- *Speech and Language Processing, 3ed*, by Daniel Jurafsky, James H. Martin <https://web.stanford.edu/~jurafsky/slp3/>
- Statquest by Josh Starmer, <https://statquest.org/video-index/>
- 3Blue1Brown, https://www.youtube.com/channel/UCYO_jab_esuFRV4b17AJtAw
- Better Explained by Kalid Azad, <https://betterexplained.com/>
- <https://www.autodraw.com/> (for some of the illustrations)



Revision of Probability and Introduction to Bayes Theorem

We are about to predict the future!



My predictions are usually very accurate

Some of you will get an **A** in this course!
and some of you will not!



Probability

Fair coin toss:

- Probability of heads: $\frac{1}{2}$ $\leftarrow$ All possible outcomes in the current universe
- Probability of tails: $\frac{1}{2}$

Fair dice roll

- Probability of getting a 6: $\frac{1}{6}$ $\leftarrow$ All possible outcomes in the current universe



Joint Probability

- Suppose we throw a white and black die simultaneously. What is the probability that the outcomes would sum to 3?
- (1,2) and (2,1) are the only two out of 36 possibilities that sum to 3.
- So: $P(\text{sums to } 3) = \frac{2}{36}$



Conditional Probability

- Now let us suppose we have already thrown the black dice and get a 2. What is the probability of “sums to 3” given this event:
- Only one possibility out of **6** possible outcomes remain
- So: $P(\text{sums to 3} \mid \text{already a 2 on black dice}) = 1/6$
(conditional probability)



From Probability to Bayes Theorem

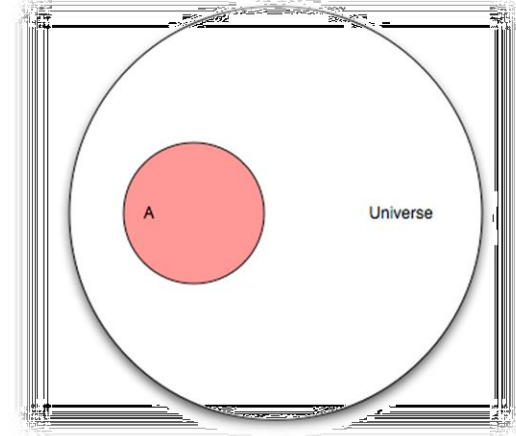
(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)

- A *Universe* with all the possible outcomes
- Interested in some subset of them (some *event*)
- Assume, we are studying cancer:
 - We observe people and see whether they have cancer or not.
 - If we take as our Universe all people participating in our study, then there are two possible outcomes for any particular individual: **either they have cancer or they do not have cancer**
 - We can then split our universe in two events:
 - the event “people with cancer” (designated as A),
 - the event “people with no cancer” (or $\neg A$)



An Event

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)



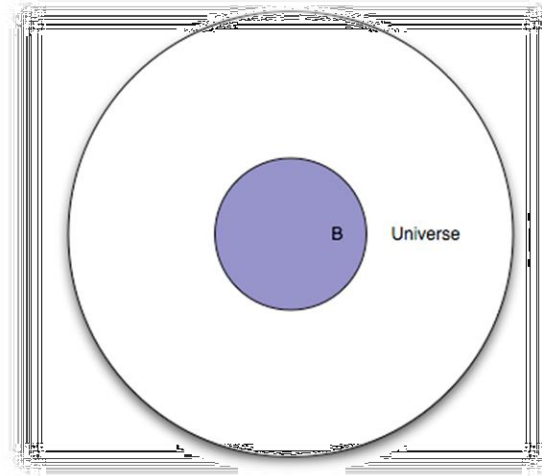
- So what is the probability that a randomly chosen person has cancer?
 - The number of elements in A divided by the number of elements of U (the Universe).
 - We denote the number of elements of A as $|A|$ (the *cardinality of A*).
 - And define the probability of A, $P(A)$, as
$$P(A) = \frac{|A|}{|U|}$$
- Since A can have at most the same number of elements as U, the probability $P(A)$ can be at most 1



Another Event

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)

- Let's say there is a new screening test that is supposed to measure *something*
- That test will be “positive” for some people, and “negative” for others
- If we take the event **B** to mean “people for whom the test is positive”.



- What is the probability that the test will be “positive” for a randomly selected person?

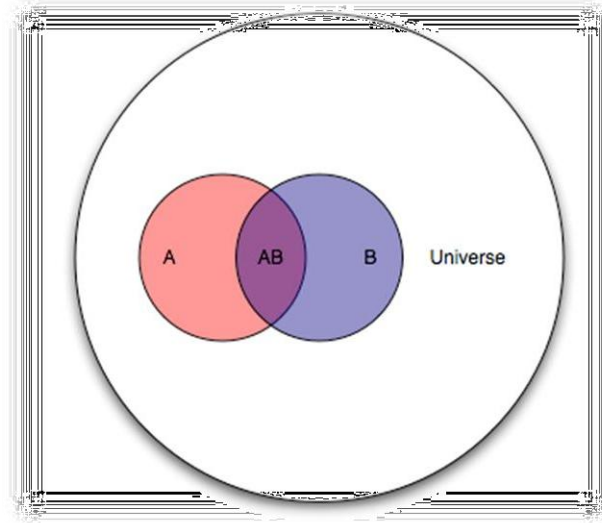
$$P(B) = \frac{|B|}{|U|}$$



The Two Events Jointly

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)

- So far, we have treated the two events in isolation.
- What happens if we put them together?



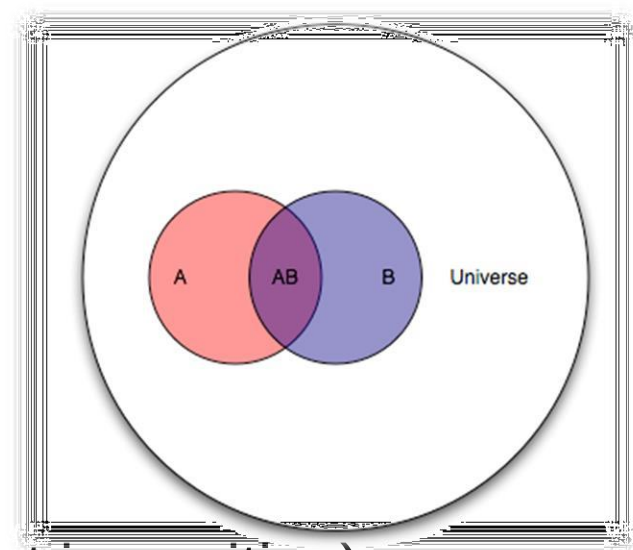
- We can compute the probability of both events occurring (**AB** is a shorthand for $\mathbf{A \cap B}$) as:

$$P(AB) = \frac{|AB|}{|U|}$$



The Two Events Jointly

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)



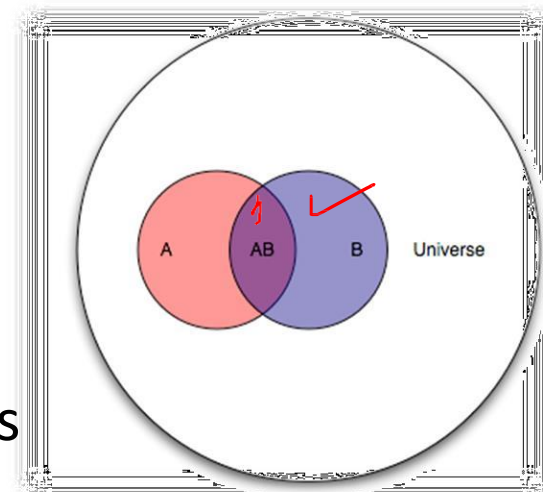
- We are dealing with:
 - an entire Universe (all people),
 - the event A (people with cancer),
 - the event B (people for whom the test is positive)
- There is also an overlap, the event AB:
 - “people with cancer and with a positive test result”.
- There is also the event B–AB:
 - “people with a positive test result and without cancer”
- And the event A–AB:
 - “people with cancer and with a negative test result”.



Conditional Probability

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)

- “**Given** that the test is positive for a randomly selected individual, what is the probability that said individual has cancer?”
- In terms of our Venn diagram:
- “Given that we are in region B, what is the probability that we are in region AB?”
- or stated differently:
- “If we make region B our new Universe, what is the probability of A?”.
- The notation for this is $P(A | B)$ (the probability of A given B



$$P(A | B) = \frac{|AB|}{|B|}$$



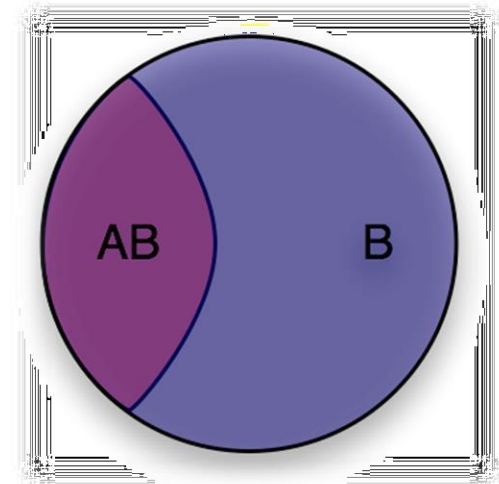
Conditional Probability

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)

- Let us convert the counts to probabilities
- Dividing both the numerator and the denominator by $|U|$, we get:

$$P(A|B) = \frac{\frac{|AB|}{|U|}}{\frac{|B|}{|U|}} = \frac{P(AB)}{P(B)} \quad (\text{Eq-1})$$

- What we've effectively done is change the Universe from U (all people), to B (people for whom the test is positive), but we are still dealing with probabilities defined in U .



The Bayes Theorem

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)

- Now let's ask the converse question:
 - “given that a randomly selected individual has cancer (event A), what is the probability that the test is positive for that individual (event AB)?”.
- It's easy to see that it is:

$$P(B|A) = \frac{P(AB)}{P(A)} \quad (\text{Eq-II})$$

- Now we have everything we need to derive Bayes' theorem, putting equations I and II together, we get

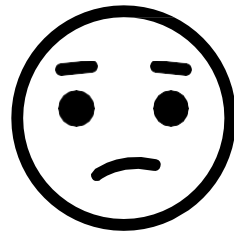
$$P(A|B)P(B) = P(B|A)P(A)$$

- which is to say $P(AB)$ is the same whether you're looking at it from the point of view of A or B, and finally

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



**But what does it
all mean?**



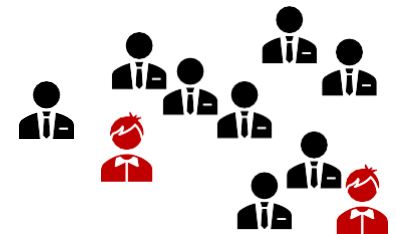
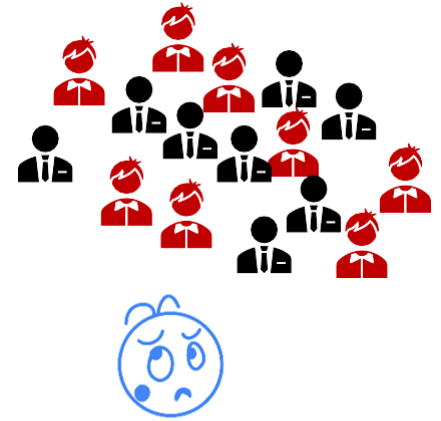
$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$



Bayes: Example 1

(Julia Galef: https://www.youtube.com/watch?v=BrK7X_XlGB8)

- You walk into a university that has Mathematics PhD and MBA students
- You run into a student who is clearly very shy
- Is he more likely to be a Mathematics PhD or an MBA?
- Mathematics PhD?
- **Hidden prior based on our belief about the state of our world:** There are always way more MBAs than Maths PhDs!



Bayes: Example 1



(Julia Galef: https://www.youtube.com/watch?v=BrK7X_XlGB8)

Assume a population of 1,100 students

- 100 are Math PhD, 1,000 are MBA
- 75% of Math (75) are shy
- 15% of Business (150) are shy
- Chances of shy student from Maths

$$\begin{aligned} P(\text{Math} | \text{Shy}) &= \frac{\text{count}(\text{Shy} \cap \text{Maths})}{\text{count}(\text{Shy})} \\ &= \frac{75}{75+150} = 33.33\% \end{aligned}$$

Note: We can easily convert the counts to probabilities by dividing the numerator and denominator by $|U| = 1,100$.



Bayes: Example 1 (directly using the Bayes Theorem)

(Julia Galef: https://www.youtube.com/watch?v=BrK7X_XlGB8)

- You walk into a university with Mathematics PhD students and MBA students
- You run into a student who is clearly very shy
- Is he more likely to be a Mathematics PhD or an MBA?

$$P(Math|Shy) = \frac{P(Shy|Math)P(Math)}{P(Shy)}$$

$$- P(Math) = \frac{Count(Math)}{\sum_{s \in subjects} Count(students\ in\ s)} = \frac{100}{1,100} = 0.0909$$

$$- P(Shy) = \frac{Count(Shy)}{\sum_{s \in subjects} Count(students\ in\ s)} = \frac{(75+100)}{1,100} = 0.2045$$

$$- P(Shy|Maths) = \frac{Count(Shy \cap Maths)}{Count(Maths)} = \frac{75}{100} = 0.75$$

Plugging in we get:

$$P(Maths|Shy) = \frac{0.75 * 0.0909}{0.2045} = \mathbf{0.3333}$$



The Bayes Theorem

Posterior

Likelihood

Prior

$$P(h|D) = \frac{P(D|h) \cdot P(h)}{P(D)}$$

Evidence or Predictive Probability

- Bayes theorem allows us to update our *Prior* $P(h)$ to the posterior $P(h|D)$ using the data
- **Prior** is our belief about the world, without looking at the data
- The **likelihood** is the probability of observing *this* particular D if a particular h were true e.g.
 - the probability of seeing a particular email assuming it's class was SPAM
 - The probability of seeing this student assuming the class was Maths PhD
- $P(D) = \sum_{h'} P(D|h')P(h')$ – The **evidence** is the probability of observing this data, D , regardless of the hypothesis (i.e. summed over all hypotheses)

The **Bayes Theorem** is a belief updating machine that relies a lot on the prior – hence different priors lead to different posteriors



MAP Hypothesis and Classification

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

Generally want the most probable hypothesis given the training data

Maximum a posteriori hypothesis h_{MAP} :

$$\begin{aligned} h_{MAP} &= \arg \max_{h \in H} P(h|D) \\ &= \arg \max_{h \in H} \frac{P(D|h)P(h)}{P(D)} \\ &= \arg \max_{h \in H} P(D|h)P(h) \end{aligned}$$

If assume $P(h_i) = P(h_j)$ then can further simplify, and choose the *Maximum likelihood* (ML) hypothesis

$$h_{ML} = \arg \max_{h_i \in H} P(D|h_i)$$

Bayes: Extension of Example 1

(Julia Galef: https://www.youtube.com/watch?v=BrK7X_XlGB8)

Now let's assume there are several subjects and students of various subjects vary on the graph of shyness

In a population of 600

- 100 Mathematicians and 75% of them are shy
- 200 Business majors and 15% of them are shy
- 300 CS majors and 50% of them are shy

Now given that a student is shy, which is the most likely subject?

- $P(\text{Maths}|\text{Shy}) = \frac{P(\text{Shy}|\text{Maths}) \cdot P(\text{Maths})}{P(\text{Shy})}$
- $P(\text{Business}|\text{Shy}) = \frac{P(\text{Shy}|\text{Business}) \cdot P(\text{Business})}{P(\text{Shy})}$
- $P(\text{CS}|\text{Shy}) = \frac{P(\text{Shy}|\text{CS}) \cdot P(\text{CS})}{P(\text{Shy})}$

And we just pick the hypothesis (subject) with the highest posterior!

- $Subject_{MAP} = \underset{Subject}{\operatorname{argmax}} \frac{P(\text{Shy}|\text{Subject}) \cdot P(\text{Subject})}{P(\text{Shy})}$

Notice that $P(\text{Shy})$ is common in the denominator and does not influence the argmax

- $Subject_{MAP} = \underset{Subject}{\operatorname{argmax}} P(\text{Shy}|\text{Subject}) \cdot P(\text{Subject})$



Bayes: Extension of Example 1

(Julia Galef: https://www.youtube.com/watch?v=BrK7X_XlGB8)

$$Subject_{MAP} = \operatorname{argmax}_{Subject} P(\text{Shy} | \text{Subject}) \cdot P(\text{Subject})$$

- $P(\text{Maths} | \text{Shy}) = 0.75 * 100/600 \rightarrow 0.125$
- $P(\text{Business} | \text{Shy}) = 0.15 * 200/600 \rightarrow 0.05$
- $P(\text{CS} | \text{Shy}) = 0.50 * 300/600 \rightarrow 0.25$

Therefore, the student is most

likely a CS major!

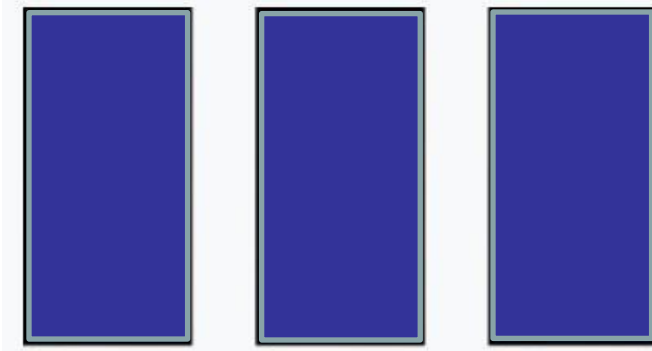
[/www.youtube.com/watch?v=BrK7X_XlGB8](https://www.youtube.com/watch?v=BrK7X_XlGB8)



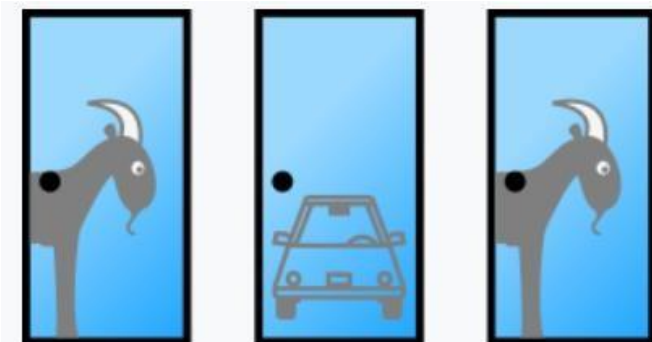
Some Warmup: Monty Hall Problem

(Khan Academy: https://www.youtube.com/watch?v=Xp6V_IO1ZKA)

- Three closed doors!



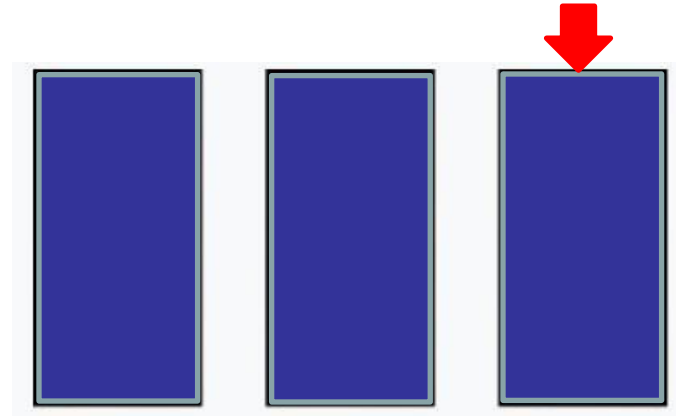
- Behind one is a great prize, while behind others are goats!



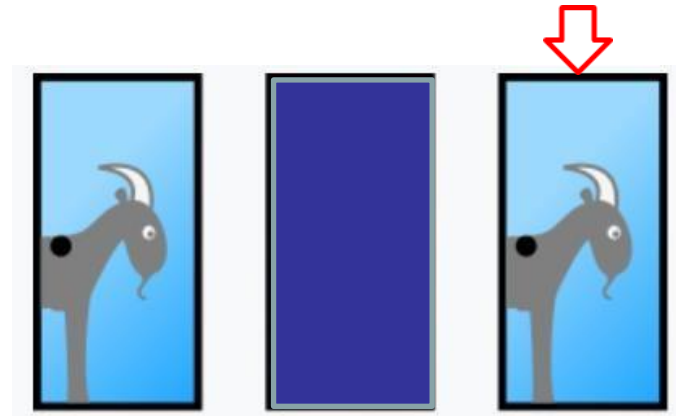
Some Warmup: Monty Hall Problem

(Khan Academy: https://www.youtube.com/watch?v=Xp6V_1O1ZKA)

- You choose a door!



- The host does not open your door but one of the other doors.



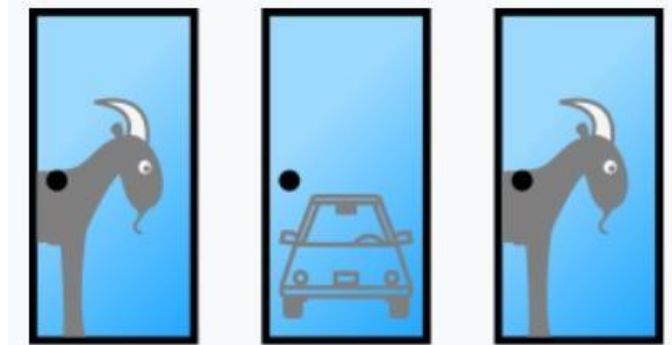
- He does not open the door with a car behind it!
- Now you are offered a chance to swap your choice.
- Would you swap? **And would it make any difference?**



(Khan Academy: https://www.youtube.com/watch?v=Xp6V_IO1ZKA)

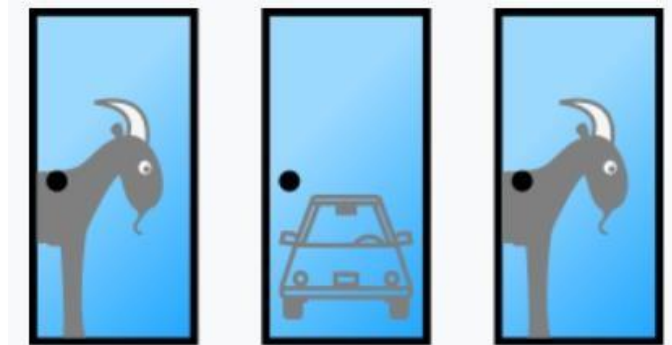
Initially,

- $P(\text{Goat}) = 2/3 = 66\%$
- $P(\text{Car}) = 1/3 = 33\%$



Now, if you don't Switch:

- $P(\text{win}) = 33\%$
- $P(\text{Fail}) = 66\%$



If you switch

- If you initially chose a goat (66% chance) and now switch, you will surely win as there is a car behind the remaining door!
- If you initially chose a car (33% chance) and now switch, you will surely lose as there is a goat behind the remaining door!
- **You should always swap!**



Behind door 1	Behind door 2	Behind door 3	Result: If staying at door #1	Result: If switching to the door offered
Goat	Goat	Car	Wins goat	Wins car
Goat	Car	Goat	Wins goat	Wins car
Car	Goat	Goat	Wins car	Wins goat

A player who stays with the initial choice wins in only one out of three of these equally likely possibilities, while a player who switches wins in two out of three.

Monty Hall Problem: A variation

(Khan Academy: https://www.youtube.com/watch?v=Xp6V_IO1ZKA)

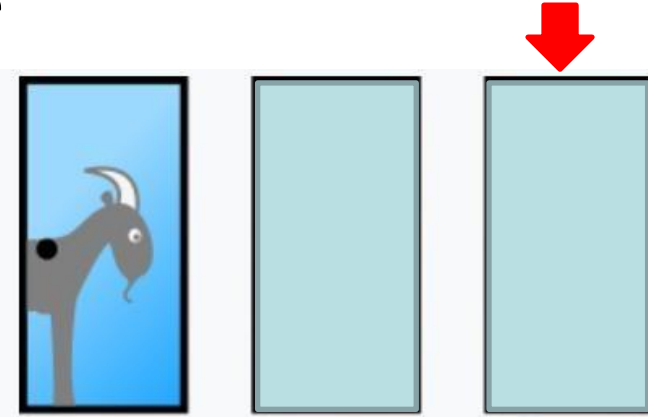
What if, instead of switching, you randomly choose again?

In the new universe you have two doors: Goat and Car

- A new random choice can yield each with 50% chance:

$$P(\text{Goat}) = 0.5$$

$$P(\text{Car}) = 0.5$$



To help convince yourselves, imagine these were not doors but boxes. One has a diamond, while two have coals.

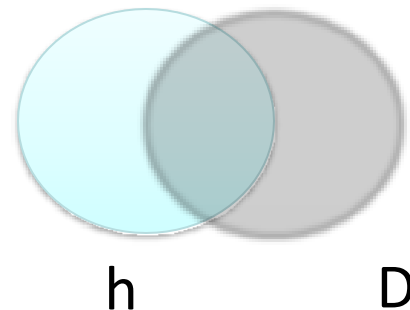
- Initially you pick a box and put it in your pocket
- The host opens one other box
- Switch means you swap the box in your pocket with the other closed box
- Re-choose means you put your box and the closed one in a bag, shuffle and pick one out.
- The old universe had two possibilities, the new one has two



Examples

In the upcoming examples we will use the following forms of conditional probability and Bayes Theorem to reach our conclusions:

- $P(h|D) = \frac{|h \cap D|}{|D|}$
- $P(h|D) = \frac{P(h \cap D)}{P(D)}$
- $P(h|D) = \frac{P(D|h) P(h)}{P(D)}$



Bayes: Example 2

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)

1% of women at age forty who participate in routine screening have cancer. 80% of women with breast cancer will get positive mammograms. 9.6% of women without cancer will also get positive mammograms.

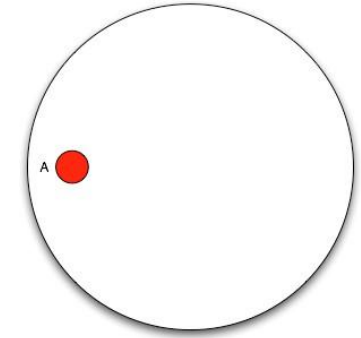
A woman in this age group had a positive mammography in a routine screening. What is the probability that she actually has cancer?



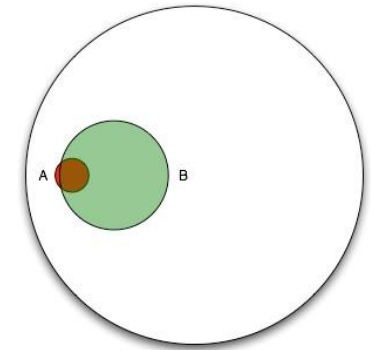
Bayes: Example 2

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)

The women with cancer, A:



The women with positive mammograms, B:

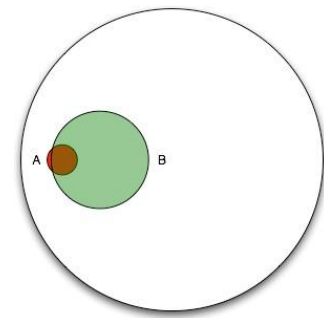


If we restrict our universe to B , only a small percentage actually have cancer.



Bayes: Example 2 – Using counts

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)



Assume we have a population of 1,000 women

1% of them (10 women) have cancer. Of these, 80% (8 women) also test positive

990 women do not have cancer. Of these, 9.6% (95 women) test positive

Of the 103 (8+95) women that test positive, only 8 have cancer.

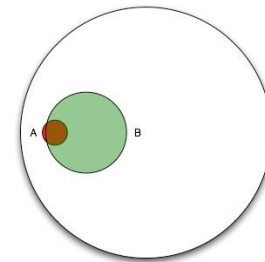
Therefore, probability of cancer given a positive result:

$$P(\text{cancer} | +) = \frac{|\text{cancer} \cap +|}{|+|} = \frac{8}{103} = 0.076$$



Bayes: Example 2 – Using h

(<https://oscarbonilla.com/2009/05/visualizing-bayes-theorem/>)



We can also do the same using MAP hypothesis:

$$\begin{aligned}P(\text{cancer}) &= 0.01 & P(\neg \text{cancer}) &= 0.99 \\P(+|\text{cancer}) &= 0.8 & P(-|\text{cancer}) &= 0.2 \\P(+|\neg \text{cancer}) &= 0.096 & P(-|\neg \text{cancer}) &= 0.904\end{aligned}$$

$$P(\text{cancer}|+) = P(+|\text{cancer}).P(\text{cancer}) = (0.8)(0.01) = 0.008$$

$$P(\neg \text{cancer}|+) = P(+|\neg \text{cancer}).P(\neg \text{cancer}) = (0.096)(0.99) = 0.095$$

Therefore, $h_{MAP} = \neg \text{cancer}$.

We can also retrieve the actual $P(\text{cancer}|+)$ by normalizing using $P(+)$.

$$\begin{aligned}P(\text{cancer}|+) &= \frac{P(+|\text{cancer}).P(\text{cancer})}{P(+)} \\&= \frac{P(+|\text{cancer}).P(\text{cancer})}{P(+|\text{cancer}).P(\text{cancer}) + P(+|\neg \text{cancer}).P(\neg \text{cancer})} \\&= \frac{0.008}{0.008+0.095} = \mathbf{0.076}\end{aligned}$$

Although H_{MAP} is $\neg \text{cancer}$, Bayes theorem has updated our belief about $P(\text{cancer})$ from 0.01 to 0.076 using data, i.e. the positive text result.



Thank you!

